BRIGHTMARTDB

SELLER & INVENTORY MANAGEMENT

Week 3 Documentation — Seller & Inventory

Purpose: This document describes the Seller and Inventory tables, their relationships, sample data, inventory analysis, and query outputs based on the submitted SQL script.

1. Seller & Inventory Table Design

Database: BrightMartDB

Seller table

Column	Data type	Constraints / purpose
Seller_ID	INT	Primary key; AUTO_INCREMENT
Seller_Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	NOT NULL; UNIQUE
Phone	VARCHAR(15)	NOT NULL; UNIQUE
Address	VARCHAR(200)	NOT NULL

Inventory table

Column	Data type	Constraints / purpose
Inventory_ID	INT	Primary key; AUTO_INCREMENT
Product_ID	INT	NOT NULL; foreign key to Product(Product_ID)
Seller_ID	INT	NOT NULL; foreign key to Seller(Seller_ID)
Stock_Quantity	INT	NOT NULL; CHECK quantity >= 0
Stock_Status	VARCHAR(20)	NOT NULL; CHECK status is Available or Out of Stock
Last_Updated	DATE	NOT NULL

Relationship: one Seller can be linked to many Inventory rows. One Product can also be linked to many Inventory rows. Inventory acts as the linking table and stores the stock quantity/status for a product supplied by a seller.

2. Updated ER Diagram

BrightMartDB — Updated ER Diagram



Seller 1 → many Inventory records; Product 1 → many Inventory records

Note: Product is referenced by the submitted SQL, but its CREATE TABLE definition is not included in that file. The Product entity is shown only with Product_ID.

3. Sample Seller & Inventory Data

Seller sample data

ID	Seller_Name	Email	Phone	Address
1	ABC Traders	abc@gmail.com	9999999999	Chennai (updated)
2	Raja Suppliers	raja@gmail.com	9876543211	Madurai
3	Sri Stores	sri@gmail.com	9876543212	Thoothukudi
4	Bright Suppliers	bright@gmail.com	9876543213	Coimbatore
5	JJ Enterprises	jj@gmail.com	9876543214	Tirunelveli
6	New Seller	new@gmail.com	9876543215	Chennai

Inventory sample data (expected after listed UPDATE)

Inventory_ID	Product_ID	Seller_ID	Qty	Status	Last_Updated
1	1	1	40	Available	2026-09-25
2	2	1	25	Available	2026-09-25
3	3	2	8	Available	2026-09-25
4	4	2	0	Out of Stock	2026-09-25
5	1	2	20	Available	2026-09-25

Assumption: the first Inventory INSERT is corrected by removing its trailing comma, all statements run in order, IDs start at 1, and DELETE Inventory_ID = 8 affects no row. Actual output can differ if the database already contained records.

4. Inventory Analysis Report

Metric	Result	Interpretation
Total inventory records	5	Five inventory rows are represented in the sample after the listed insert and update.
Available records	4	Four rows have Stock_Status = Available.
Out-of-stock records	1	Product_ID 4 has quantity 0 and status Out of Stock.
Low stock (< 10)	2	Product_ID 3 has 8 units; Product_ID 4 has 0 units.
Total units in stock	93	40 + 25 + 8 + 0 + 20 = 93 units.
Average quantity per inventory row	18.6	93 units divided by 5 inventory rows.

Metric	Result	Interpretation
Highest stock	40 units	Product_ID 1 after the update.
Seller with most inventory rows	Raja Suppliers	Three inventory rows are linked to Seller_ID 2.

Recommendations: replenish Product_ID 4 first because it is out of stock; review Product_ID 3 because it has fewer than 10 units; update Last_Updated whenever stock changes.

5. Query Output Screenshots / Output Samples

The submitted file contains SQL queries but no actual database screenshots. The tables below are illustrative output samples calculated from the intended sample data; they are not captured screenshots from a running MySQL session.

5.1 Seller-wise inventory

Seller_Name	Product_ID	Stock_Quantity	Stock_Status
ABC Traders	1	40	Available
ABC Traders	2	25	Available
Raja Suppliers	3	8	Available
Raja Suppliers	4	0	Out of Stock
Raja Suppliers	1	20	Available

5.2 Inventory by stock status

Product_ID	Stock_Quantity	Stock_Status
1	40	Available
2	25	Available
3	8	Available
4	0	Out of Stock
1	20	Available

5.3 Stock summary queries

SQL analysis	Expected result
COUNT where Stock_Status = 'Available'	4
COUNT where Stock_Status = 'Out of Stock'	1
ORDER BY Stock_Quantity DESC LIMIT 1	Product_ID 1, Stock_Quantity 40
AVG(Stock_Quantity)	18.6

Important SQL execution notes

• The first Inventory INSERT ends with a comma after the final value row. Remove that comma before executing the script.

• The Inventory table has a foreign key to Product(Product_ID), but the uploaded script does not create or populate Product. Create/populate Product first, with Product_ID values 1–4, before inserting Inventory rows.

• The query DELETE FROM Inventory WHERE Inventory_ID = 8 will only delete a row if Inventory_ID 8 exists; under the sample sequence shown here it does not.

• These output samples assume the Seller and Inventory inserts succeed in a clean database. Run the corrected SQL in MySQL Workbench and replace the illustrative tables with real screenshots if your instructor specifically requires screenshots.

Conclusion

The Seller and Inventory design records seller contact information and tracks product stock by seller. The sample analysis shows 93 total units across five inventory records, with one out-of-stock product and one additional low-stock product requiring attention.